

1 Solution — GIAM 1.1

Problem 1

1.1 Problem

Each of the quantities indexing the rows of the following table is in one or more of the sets which index the columns. Place a check mark in a table entry if the quantity is in the set.

The quantities are $17, \pi, 22/7, -6, e^0, 1+i, \sqrt{3}, i^2$; the sets are $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

1.2 Convention — What is $\mathbb{N}$?

Section 1.1 fixes the *traditional* convention:

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}.$$

So $\mathbb{N}$ contains the positive integers — **no zero, no negatives**. The other inclusions are the usual

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

Every number that lies in a set automatically lies in all of the larger ones — so a check in $\mathbb{N}$ implies checks in all of $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

1.3 Reducing the Two Mystery Entries

Two of the rows are not given in “obviously classifiable” form:

- $e^0 = 1$. (Any nonzero base to the zero power is 1.)
- $i^2 = -1$. (By the defining property of i .)

So the row " e^0 " is the integer 1, and the row " i^2 " is the integer -1 .

1.4 The Filled Table

Quantity	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{C}$
17	✓	✓	✓	✓	✓
π				✓	✓
$22/7$			✓	✓	✓
-6		✓	✓	✓	✓
e^0	✓	✓	✓	✓	✓
$1 + i$					✓
$\sqrt{3}$				✓	✓
i^2		✓	✓	✓	✓

Table 1.1

1.5 Row-by-Row Justification

- **17.** A positive integer ≥ 1 , so in $\mathbb{N}$ and hence in everything.
- **π .** Irrational (a fact we accept; it'll be proved later). Therefore not in $\mathbb{Q}$ (and so not in $\mathbb{Z}$ or $\mathbb{N}$). It *is* a real number, so it's in $\mathbb{R}$ and $\mathbb{C}$.
- **$22/7$.** A ratio of two integers with nonzero denominator, so in $\mathbb{Q}$. But $22 = 3 \cdot 7 + 1$, so $22/7 = 3 + 1/7 \notin \mathbb{Z}$. Hence in $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ only.
- **-6 .** A negative integer. In $\mathbb{Z}$ — but *not* in $\mathbb{N}$ under the chosen convention. So in $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.
- **$e^0 = 1$.** A positive integer ≥ 1 . In every set.
- **$1 + i$.** Has imaginary part $1 \neq 0$, so *not* in $\mathbb{R}$ (and so not in $\mathbb{Q}, \mathbb{Z}, \mathbb{N}$). It *is* by definition in $\mathbb{C}$.
- **$\sqrt{3}$.** Irrational (since **3** isn't a perfect square — short proof in the remarks). So in $\mathbb{R}, \mathbb{C}$ only.
- **$i^2 = -1$.** A negative integer. Like -6 : in $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

1.6 Visual — Nesting the Eight Quantities

Each number lives in a *smallest* containing set; the bigger sets follow for free. The diagram below places each entry inside the smallest set that contains it (and therefore inside every set surrounding that one).

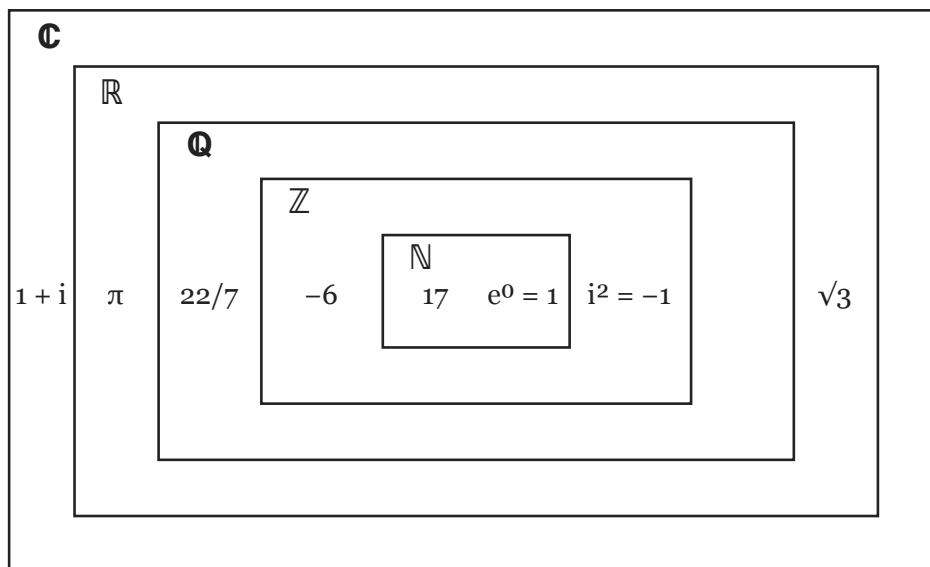


Figure 1.1: Each quantity is shown in the innermost set containing it. Items inside $\mathbb{N}$ are also in $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ — and so on outward.

Reading the figure:

- **Innermost (in $\mathbb{N}$):** 17 and $e^0 = 1$.
- **In $\mathbb{Z} \setminus \mathbb{N}$:** -6 and $i^2 = -1$ (the negative integers).
- **In $\mathbb{Q} \setminus \mathbb{Z}$:** $22/7$ (a non-integer rational).
- **In $\mathbb{R} \setminus \mathbb{Q}$:** π and $\sqrt{3}$ (irrationals).
- **In $\mathbb{C} \setminus \mathbb{R}$:** $1 + i$ (genuinely complex).

The number of check marks for each row is just *how many sets surround its level*: 5 for items in $\mathbb{N}$, 4 for items in $\mathbb{Z} \setminus \mathbb{N}$, 3 for items in $\mathbb{Q} \setminus \mathbb{Z}$, 2 for items in $\mathbb{R} \setminus \mathbb{Q}$, 1 for items in $\mathbb{C} \setminus \mathbb{R}$.

1.7 Remark — Why π and $\sqrt{3}$ are Irrational

The textbook accepts these as facts at this point in the book. Both have short proofs:

- $\sqrt{3} \notin \mathbb{Q}$. Suppose $\sqrt{3} = a/b$ in lowest terms with a, b integers, $b > 0$. Squaring gives $3b^2 = a^2$. So $3 \mid a^2$, hence $3 \mid a$ (since 3 is prime). Write $a = 3a'$; then $3b^2 = 9a'^2$, i.e. $b^2 = 3a'^2$. By the same argument $3 \mid b$. But then 3 divides both a and b , contradicting “lowest terms”. So no such (a, b) exists. ■
- $\pi \notin \mathbb{Q}$. This is *much* harder; the standard proof (Niven, 1947) uses calculus and definite integrals. Out of scope for Chapter 1, but the result is famous enough to take on faith.

1.8 Remark — $22/7$ is Not π

$22/7 \approx 3.1428571\dots$ is a *rational* approximation to π , popularised by Archimedes. The two are *different numbers* — $22/7$ sits in $\mathbb{Q}$, while π does not. In fact $22/7 - \pi \approx 0.00126 > 0$, so $22/7$ is slightly *larger* than π .

A common slip: assuming “the rational approximation $22/7$ ” is the same row as “ π ”. The exercise tests precisely whether the student keeps them straight.

1.9 Remark — Why $1 + i \notin \mathbb{R}$

The real numbers $\mathbb{R}$ are exactly the complex numbers with zero imaginary part. $1 + i$ has imaginary part $1 \neq 0$, so it is *not* real. The cleanest way to see it: if $1 + i$ were real, then $i = (1 + i) - 1$ would also be real — but $i^2 = -1 < 0$, and no real number has a negative square. Contradiction.

1.10 Remark — Compactly: Which Set is “Smallest”?

The cleanest summary of the answer is a single column listing the *smallest* containing set for each quantity. The full row of checks is then determined: tick that set plus every set to its right in the chain $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.

Quantity	Smallest set	Check count
17	$\mathbb{N}$	5

$e^0 = 1$	$\mathbb{N}$	5
-6	$\mathbb{Z}$	4
$i^2 = -1$	$\mathbb{Z}$	4
$22/7$	$\mathbb{Q}$	3
π	$\mathbb{R}$	2
$\sqrt{3}$	$\mathbb{R}$	2
$1 + i$	$\mathbb{C}$	1

Table 1.2

This is the structural way to read the answer table — the chain of inclusions does all the bookkeeping for free.

1.11 Remark — Where the Convention $0 \notin \mathbb{N}$ Matters Here

It doesn't, for *this* exercise — none of the eight rows is the number 0 . If the row " $\sin(0)$ " or " $0! - 1$ " appeared, the convention would decide whether to check the $\mathbb{N}$ column for it. Under the textbook's $\mathbb{N} = \{1, 2, 3, \dots\}$: no. Under the other convention $\mathbb{N} = \{0, 1, 2, \dots\}$: yes.

Worth flagging because in many other courses — and in most computer-science contexts — $0 \in \mathbb{N}$ is standard. The textbook's choice is just one of two equally common conventions, and it's worth being aware of which one your current setting uses.